

A top-down view of a dark desk. In the top right corner, there is a white cup. Next to it is a silver laptop with a black keyboard and a black pen resting on its trackpad. In the bottom right corner, a pair of black-rimmed glasses is lying on the desk. The background is a dark, textured surface.

Database Management Systems

Freelancer Marketplace Database Project

Sushmitha Katherine Jayaraj
Jayaraj.s@northeastern.edu

Project Overview

Goal: To design a relational database for a freelancer marketplace to manage users, jobs, bids, applications, and reviews.

Key Features: User roles (clients and freelancers).

- Job management and bidding system.
- Reviews and ratings for completed jobs.

Database Highlights

- **Tables:** Users, Jobs, Applications, Bids, Skills, Reviews and Job skills

Relationships:

- One client can post multiple jobs.
- A freelancer can apply or bid for multiple jobs.
- Jobs are linked to reviews once completed.

Personas

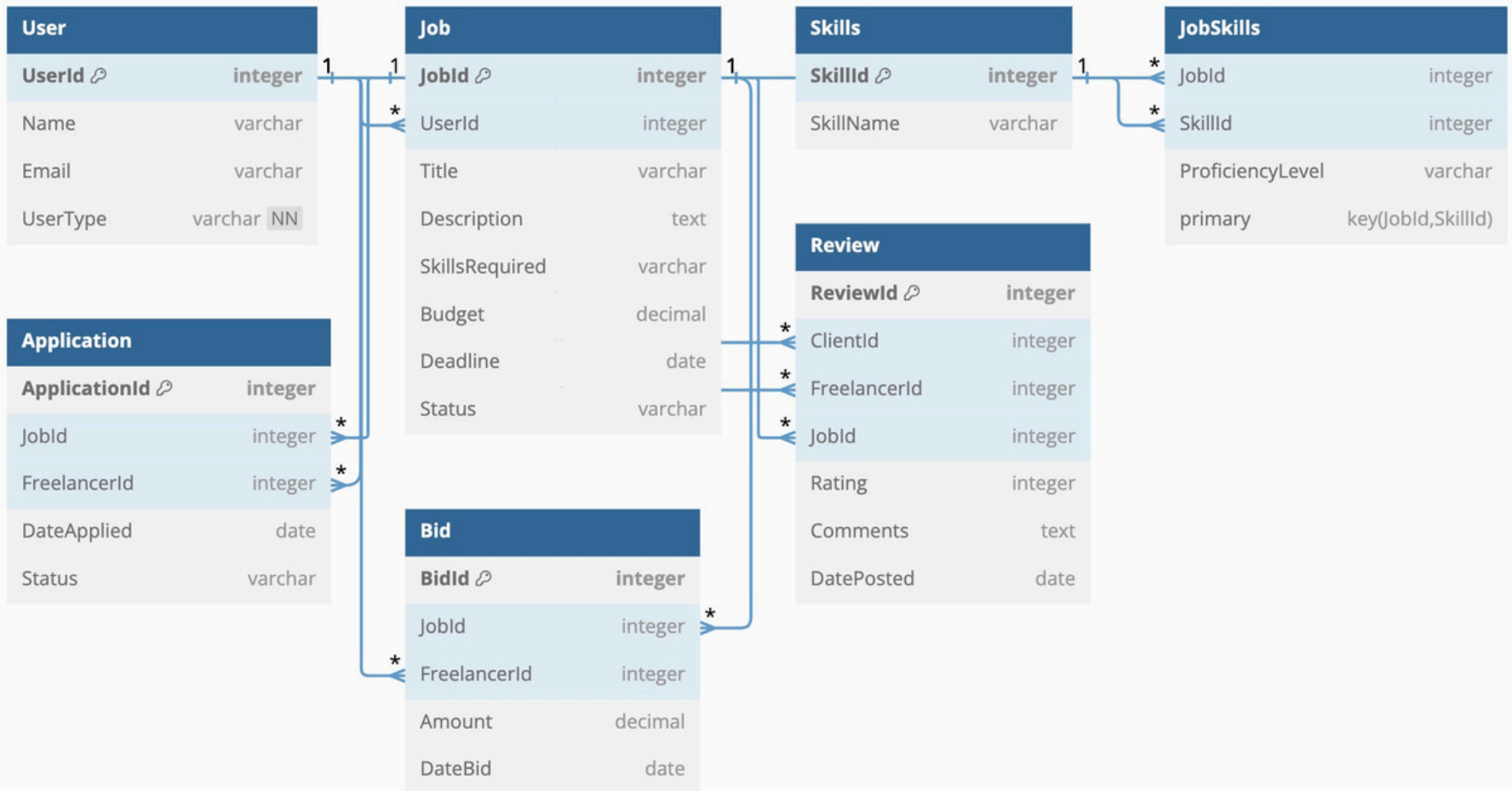
Persona 1: Client

- **Who:** A user posting tasks they need completed (e.g., Sarah, small business owner).
- **Use Cases:**
 - Post jobs with skills, budget, and deadlines.
 - Review and select freelancer applications or bids.
 - Approve completed work and leave feedback.

Persona 2: Freelancer

- **Who:** A user seeking job opportunities to earn money (e.g., Jason, web developer).
- **Use Cases:**
 - Search and bid on jobs based on skills.
 - Apply for jobs with portfolios and qualifications.
 - Deliver tasks and earn reviews.

ER Diagram



Business Rules & Facts

1. User Types: Users are either Clients (post jobs) or Freelancers (apply or bid for jobs).
2. Job Posting: Only Clients can post jobs with required skills, budget, and deadlines.
3. Bidding & Applications: Freelancers can apply or bid for jobs, and Clients review them to select candidates.
4. Review System: Clients leave ratings and reviews for Freelancers after job completion.
5. Scope Exclusions: No direct messaging, payment integration, or file uploads in the current version.

SQL Examples

1. Total Bids per Job

```
SELECT Job.Title, COUNT(Bid.BidId) AS TotalBids
FROM Job
JOIN Bid ON Job.JobId = Bid.JobId
GROUP BY Job.JobId;
```

	Title	TotalBids
1	Logo Design	8
2	Website Creation	6
3	HTML Email Template	7

2. Retrieve all jobs with their client's name and budget

```
SELECT Job.Title, User.Name AS Client, Job.Budget
FROM Job
JOIN User ON Job.UserId = User.UserId
WHERE Job.Status = 'Open';
```

	Title	Client	Budget
1	Logo Design	Sarah	100.0
2	Website Creation	Anna	800.0
3	HTML Email Template	Sarah	150.0

3. Listing all jobs with the lowest bid and freelancer name

```
SELECT Job.Title, MIN(Bid.Amount) AS LowestBid, User.Name AS Freelancer
FROM Job
JOIN Bid ON Job.JobId = Bid.JobId
JOIN User ON Bid.FreelancerId = User.UserId
GROUP BY Job.JobId;
```

	Title	LowestBid	Freelancer
1	Logo Design	85.0	Jason
2	Website Creation	730.0	Anna
3	HTML Email Template	130.0	Sarah

4. Average rating of freelancers who received reviews

```
SELECT User.Name AS Freelancer, AVG(Review.Rating) AS AvgRating
FROM Review
JOIN User ON Review.FreelancerId = User.UserId
GROUP BY User.UserId;
```

	Freelancer	AvgRating
1	Jason	5.0
2	Mike	4.0

5. Freelancers who applied for each job with the application status

```
SELECT Job.Title, User.Name AS Freelancer, Application.Status
FROM Application
JOIN User ON Application.FreelancerId = User.UserId
JOIN Job ON Application.JobId = Job.JobId;
```

	Title	Freelancer	Status
1	Logo Design	Jason	Pending
2	Website Creation	Mike	Pending
3	HTML Email Template	Jason	Pending

6. Jobs with the freelancers' reviews and ratings

```
SELECT Job.Title, User.Name AS Freelancer, Review.Rating, Review.Comments
FROM Review
JOIN User ON Review.FreelancerId = User.UserId
JOIN Job ON Review.JobId = Job.JobId;
```

	Title	Freelancer	Rating	Comments
1	Logo Design	Jason	5	Excellent work!
2	Website Creation	Mike	4	Very good job on the website!

Metrics and Analytics

1. Clients

- Number of Jobs Posted: Total jobs posted.
- Number of Bids per Job: Average bids per job.
- Job Success Rate: Percentage of completed jobs.

2. Freelancers

- Number of Bids Submitted: Total bids submitted.
- Acceptance Rate: Percentage of accepted bids.
- Average Rating: Average review rating from clients.

3. Administrators

- Number of Active Users: Clients and freelancers with active profiles.
- User Retention: Percentage of returning users.
- Dispute Resolution Rate: Percentage of disputes resolved

Security & Privacy Aspects

- 1.Sensitive Data: User information (emails, profiles), job details, bid amounts, and reviews are stored. Freelancers' payment details are managed externally.
- 2.Target by Hackers: Personal information, payment history, and reviews could be targeted for phishing attacks, identity theft, or fraud.
- 3.Data Encryption: Ensure that sensitive data (user info, job details, and payment records) is encrypted during storage and transmission.
- 4.Access Control: Admins and users should have role-based access to ensure only authorized individuals can access sensitive data (e.g., freelancers' earnings or client details)

Next Steps / Lessons Learned

- 1.Database Design & Integrity: Structuring the Freelancer Marketplace database with well-defined relationships (e.g., Job-Bid, Client-Freelancer) ensures data consistency and scalability.
- 2.Query Optimization: Efficient JOIN queries improve performance, enabling fast retrieval of job, bid, and user data.
- 3.Future Improvements: Enhance the user interface for better user experience and integrate a secure payment gateway for handling transactions directly on the platform.

